

Dr. M. Nur Hasan

Computational Materials Physicist

(Electronic Structure, Magnetism and Multiscale Modelling)

Materials Theory, Department of Physics and Astronomy

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Professional Profile

Computational materials physicist specializing in first-principles electronic-structure theory and multiscale modelling of magnetic materials. Expertise includes DFT, exchange interactions, magnetocrystalline anisotropy, finite-temperature phenomena, and atomistic spin dynamics, with applications to permanent magnets, quantum materials, and low-dimensional systems. Currently a Postdoctoral Researcher at Uppsala University within the EU Horizon Europe MaMMoS project.

Research Expertise

Electronic structure theory • Computational magnetism • Permanent magnets • Magnetic anisotropy • Exchange interactions • Finite-temperature magnetism • Atomistic spin dynamics • 2D and van der Waals magnetism • Spintronics and chiral magnetism • Correlated quantum materials • Multiscale materials modelling.

Professional Experience

Postdoctoral Researcher

[April 2024 – Present]

Materials Theory, Department of Physics and Astronomy, Uppsala University, Sweden
EU Horizon Europe MaMMoS Project

First-principles modelling of rare-earth-free permanent magnets and magnetic inter-metallic compounds; calculation of magnetic anisotropy, exchange interactions, magnetization and finite-temperature magnetic properties; atomistic spin-dynamics simulations using UppASD; multiscale connection between electronic-structure parameters and permanent-magnet performance; high-throughput computational screening and chemical-disorder modelling; validation of machine-learning models for Curie temperature and saturation magnetization within MaMMoS.

PhD Researcher (CSIR fellow)

[2017–2023]

S. N. Bose National Centre for Basic Sciences, Kolkata, India

First-principles investigation of correlated and magnetic quantum materials, including kagome metals, van der Waals magnets and spintronic systems. Research resulted in publications in Physical Review Letters, Physical Review B, Nano Letters, npj 2D Materials and Applications and related journals.

Selected Research Highlights

Rare-Earth-Free Permanent Magnets: First-principles and multiscale design of Y–Fe-based permanent magnets, focusing on chemical substitution and disorder, magnetocrystalline anisotropy, exchange interactions, Curie temperature, atomistic spin dynamics, and micromagnetic performance. This work forms part of the EU Horizon Europe MaMMoS project at Uppsala University.

Two-Dimensional and van der Waals Magnetism: Investigation of low-dimensional magnetic materials, particularly Fe_4GeTe_2 and 1T-CrS_2 , addressing electronic correlations, spin–orbit coupling, magnetic exchange, anisotropy, spin reorientation, spin–

phonon coupling, and finite-temperature magnetism using first-principles, DFT+DMFT, and atomistic spin-dynamics approaches. *Phys. Rev. B* **111**, 134449 (2025); *ACS Nano* **20**, 24621 (2026); *arXiv:2607.23625* (2026).

Spintronics and Chiral Magnetism: First-principles investigation of interfacial Dzyaloshinskii–Moriya interaction, spin–orbit-driven phenomena, and chiral magnetism in transition-metal dichalcogenide/ferromagnet heterostructures and magnetic interfaces. *ACS Nano* **19**, 23564–23574 (2025).

Correlated Quantum Materials: Investigation of correlation-driven magnetism in kagome AV_3Sb_5 ($\text{A} = \text{K}, \text{Rb}, \text{Cs}$), including dynamical electronic correlations, magnetic fluctuations, exchange interactions, and complex magnetic textures. *Phys. Rev. Lett.* **131**, 196702 (2023); *Phys. Rev. B* **108**, 174413 (2023).

Technical Expertise

Electronic Structure: Density functional theory(DFT), DFT+ U , spin–orbit coupling, DFT+DMFT, correlated electronic-structure methods, electronic bands, density of states, Fermi-surface analysis, and high-throughput first-principles calculations.

Computational Magnetism: Magnetocrystalline anisotropy, Magnetic exchange interactions, Dzyaloshinskii–Moriya interaction, anisotropic exchange, finite-temperature magnetism, Monte Carlo simulations, atomistic spin dynamics, spin textures and chiral magnetism.

Materials Modelling: Chemical disorder and SQS, structural optimization, phonons, vibrational free energies, phase stability, finite-temperature materials modelling, machine-learning interatomic potentials (MLIPs), and multiscale magnetic-materials design.

Software: VASP, RSPt, UppASD, ATAT, Phonopy, MatterSim, Wannier90, WannierTools, WannierBerri, ASE, IFermi, PyProcar, FermiSurfer, VESTA, and XCrySDen.

Programming & HPC: Python, Bash/Shell scripting, workflow automation, SLURM, MPI, and large-scale parallel computing on national and European HPC systems including Dardel, Tetralith, LUMI, Pelle/UPPMAX, and Arrhenius.

Selected Publications

R. Pal, **M. N. Hasan**, D. Das, C. Nayak, M. Deka, N. Salehi, M. Pereiro, S. Mondal, A. Misra, A. Singha, P. Mandal, D. Karmakar, A. N. Pal; *Spin Reorientation Driven Renormalization of Spin–Phonon Coupling in Fe_4GeTe_2* . *ACS Nano* **20**, 24621 (2026).

M. N. Hasan, N. Salehi, F. Sorgenfrei, A. Delin, I. Di Marco, A. Bergman, M. Pereiro, P. Thunström, O. Eriksson, & D. Karmakar; Dynamical electronic correlations and chiral magnetism in the van der Waals magnet Fe_4GeTe_2 . *Phys. Rev. B* **111**, 134449 (2025).

S. Pal, **M. N. Hasan**, H. Bangar, N. Salehi, S. Mathur, M. Pereiro, P. Thunström, P. K. Muduli, D. Karmakar, & A. Barman; Deciphering Interfacial Dzyaloshinskii–Moriya Interaction in Transition-Metal Dichalcogenide/Permalloy Heterostructures by Brillouin Light Scattering and First-Principles Calculations. *ACS Nano* **19**, 23564–23574 (2025).

M. N. Hasan, R. Bharati, J. Hellsvik, A. Delin, S. K. Pal, A. Bergman, S. Sharma, I. Di Marco, M. Pereiro, P. Thunström, P. M. Oppeneer, O. Eriksson, & D. Karmakar; Magnetism in AV_3Sb_5 (Cs , Rb and K): Origin and Consequences for the Strongly Correlated Phases. *Phys. Rev. Lett.* **131**, 196702 (2023).

D. Karmakar, M. Pereiro, **M. N. Hasan**, R. Bharati, J. Hellsvik, A. Delin, S. K. Pal, A. Bergman, S. Sharma, I. Di Marco, P. Thunström, P. M. Oppeneer, & O. Eriksson; Magnetism in AV_3Sb_5 : Complex Texture of the Dynamical Magnetic Properties. *Phys. Rev. B* **108**, 174413 (2023).

Total Publications: 33+ peer-reviewed— Citations: 620+ — h-index: 12

Full publication list and citation metrics available via [\[Google Scholar\]](#)

Research Funding & Computational Allocations	UPPMAX Computational Allocation, Sweden [2026] Awarded computational resources through an independently submitted research proposal for first-principles and computational-magnetism simulations.
	NAISS / Arrhenius Computational Allocation, Sweden [2026] Awarded computational resources through an independently submitted research proposal for large-scale electronic-structure and magnetic-materials simulations on the Arrhenius HPC system.
Academic Service & Scientific Activities	Co-organizer, European School on Magnetism (ESM 2026) [2026] <i>Spin-orbit driven magnetic phenomena</i> , Uppsala University, Uppsala, Sweden.
	Co-organizer Oral Presenter, DQMMA 2023 [2023] First Indo–Swedish Meeting on <i>Divergent Quantum Materials, Methods and Applications</i> , Goa, India. Contributed to the scientific organization of the meeting and delivered an oral presentation.
Education	Doctor of Philosophy (Ph.D.), Physics [December 2023] S. N. Bose National Centre for Basic Sciences, Kolkata, India Awarded by the University of Calcutta, India.
	Master of Science (M.Sc.), Physics [2015 - 2017] Department of Physics, Aligarh Muslim University, Aligarh - 202002, India.
	Bachelor of Science in Physics [2012 - 2015] Department of Physics, Aligarh Muslim University, Aligarh - 202002, India.
Awards & Fellowships	SERB International Travel Grant [2022]
	CSIR Foreign Travel Grant [2022]
	CSIR Senior Research Fellowship (SRF) [2020–2023]
	CSIR Junior Research Fellowship (JRF) [2018–2020]
	CSIR-NET Junior Research Fellowship [2017]
	Joint Entrance Screening Test (JEST) [2018]
Languages	Graduate Aptitude Test in Engineering (GATE), Physics [2017, 2018]
	English: Speak, Read, Write
	Bengali: Speak, Read, Write (Mother tongue)
	Hindi: Speak, Read (Working proficiency)